

Machine Learning for American Option Pricing and Option-Chain Microstructure: Benchmark Trading Against the Black–Scholes Surface on Synthetic and Real-World SPY Data

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Abstract—The paper explores the question of whether supervised machine learning (ML) can be applied as a suitable and better estimate for American call values compared to a classical Black–Scholes (BS) benchmark, and whether implied volatility (IV) and bid-ask spreads can be predicted variably using listed quotes by using machine learning. A 29,671-row high-step binomial American-call dataset is cleaned on the synthetic track and stratified by moneyness and maturity, which are thereafter sampled to 4,998 streamlined contracts. Fifteen theory-motivated features such as a Black–Scholes European-call benchmark are fed on six regressors (Linear, Ridge, Random Forest, Gradient Boosting, Support Vector Regression and a Multi-Layer Perceptron). Gradient Boosting achieves the lowest hold-out error (MAE = 0.210, RMSE = 0.323, $R^2 = 0.9992$); a 5-fold cross-validated SVR attains the lowest CV-MAE (0.0954 ± 0.0019). Permutation importance reveals that the BS European price is approximately 40 times the explanatory power of any other feature, closed-form structure captures most of the variation and ML learns the early-exercise premium. On the empirical track, 8,722 SPY quotes from Yahoo Finance (32 expiries; 8,063 after cleaning) are modelled: SVR predicts IV with $R^2 = 0.981$ (MAE = 0.014), an MLP predicts spread with $R^2 = 0.994$ (MAE = 0.057); linear baselines collapse to $R^2 \approx 0.61$ and 0.64. ML does not replace but adds to the BS surface on synthetic data and significantly outperforms linear baselines on microstructure-contaminated real data. Feature leakage, single-snapshot sampling, and a constant risk-free rate are flagged as considerable limitations.

Index Terms—American option pricing, Black–Scholes, binomial tree, gradient boosting, implied volatility, bid–ask spread, market microstructure, permutation importance, support vector regression, SPY option chain.

I. INTRODUCTION

One of the most difficult problems in quantitative finance is the American-style derivatives since the pricing problem is a free-boundary Partial Differential Equation (PDE) with no general closed-form solution [1], [2]. Valuation schemes used in industry standard are based on numerical schemes, lattice methods, Least Squares Monte Carlo, finite-difference PDE solvers, which are based on trade-offs between accuracy and runtime. The growth in option books, prime-brokerage exposures and retail flow since 2008 has augmented the marginal worth of quick and precise American pricers [3], [4].

There are three forces that drive an ML reappraisal. Deep learning and gradient-boosted ensembles provide flexible func-

tion approximators, and prior research reveals their computational efficiency relative to traditional numerical methods, mainly at inference time [5], [7]. High-step binomial trees provide synthetic datasets that provide a controlled laboratory with ground truth that is analytically defined. Market microstructure effects, including dealer inventory, order-flow imbalances, and stale quotes, affect implied volatility and bid-ask spreads and are not explicitly modelled by closed-form models and drive the adoption of data-driven models [7]. Fundamentally, these three forces are the maturity of flexible function approximators, the availability of controlled synthetic datasets, and the microstructure-driven nature of empirical option chain, which are relatable when considering capable models, clean training data and real-world quantities that are uncapturable by classical theories.

Therefore, there are three things that this paper meticulously contributes. First, a theory-conscious pipeline that incorporates the Black–Scholes European call price as an explicit feature, with permutation importance used to break down the pricing error, a diagnostic design that tests whether ML learns the early-exercise premium or simply relearns closed-form structure. Second, a similar comparison of six regressor families on a 4,998-row stratified synthetic sample and an 8,063-row SPY snapshot of the same sample with the same preprocessing. Third, an interpretability layer, 5-fold cross-validated hyperparameter search, residual stratification, paired t-tests of forecast errors, and a microstructure-based reading of IV and spread predictability.

Although research has been done on ML option pricing [8], implied volatility with transformers [9], and bid-ask spreads [10], not many benchmark ML pricers on synthetic data or extend the analysis to real-time SPY chains. This paper compares six regressors on both, and tests permutation importance to determine whether ML augments or simply relearns the Black–Scholes surface.

Research Questions

The following research questions constitute the basis for this study:

RQ1: Does ML regressors outperform a Black–Scholes European model in estimating American calls, and to what

extent is the outperformance due to the early-exercise premium or to the closed-form component?

RQ2: To what extent can ML regressors predict IV and bid-ask spreads on actual SPY quotes, and does the linear-nonlinear gap indicate microstructure curvature that cannot be seen by closed-form models?

II. RELATED WORK

A. Classical American Option Pricing

Black-Scholes-Merton model [11], [12] admit closed form in the case of European exercise. The binomial tree of Cox et al. [1] provided the first scheme that was tractable with respect to the American exercise boundary and remains the academic standard. Longstaff and Schwartz [2] introduced Least Squares Monte Carlo. Hull [3] observes that when the stock is not paying dividends, the American call premium is zero and therefore the American and European prices coincide. The premium is strictly positive and state-dependent under positive dividend yield q , as in the synthetic dataset that is analysed in this paper. Recent ML solutions to the American problem, specifically, Chen et al. [13] under stochastic volatility and Goudenège et al. [14] under high-dimensional Markovian and non-Markovian models, show that neural networks can be more effective than classical LSM in realistic regimes.

B. Machine Learning in Option Pricing

ML option pricing has evolved beyond the initial neural-network pricers trained to the Black-Scholes surface to production-quality deep calibration systems. A survey of over fifty of such studies by Ruf and Wang [16] find that ML pricers are seldom accurate than classical schemes, but can provide microsecond inferences once trained, compared to milliseconds with a 1,000-step binomial tree.

Although Ruf and Wang [16] are correct that ML is faster and less accurate, but Kim et al. [15] demonstrate that in a real-world trading setting, the inaccuracy of ML is its strength, as it can see profitable opportunities that perfectly accurate classical models are oblivious to.

Buehler et al. [17] proposed deep hedging; a paradigm where pricing and hedging problem is learned end-to-end. Incrementally, Liu et al. [18] further provided a production-scale calibration framework. Also, Richter and Teichmann [19] incorporate theoretical consistency constraints into the network, generalising previous homogeneity-hint methods, by deep calibration. While the Deep Galerkin Method (DGM) and the backward stochastic differential equation (BSDE) approaches are scalable [20], the SHoP-BS neural PDE framework employs matrix-based derivatives to obtain efficient, physically consistent pricing and Greeks [21].

Kohler et al. [22] proved convergence of NN-based variants of LSM; Becker et al. [23] established deep optimal stopping in high-dimensional Bermudan baskets. LightGBM and XGBoost are the leaders in tabular finance-tree ensembles [24], [25]. Recent research by Pawar and Chaudhary [8] extends this lineage of transformer based architectures by fine-tuning GPT-3.5 turbo on five-years of AAPL, GOOG, and MSFT

option-chain data and reveals consistent MAE and RMSE improvements over Black-Scholes, with ATM-options priced accurately and ITM-options consistently underpriced, which is a stratified error pattern that reflects the innate-ITM long-dated tail observed in Fig. 4 within this paper.

C. Implied Volatility and Bid-Ask Spread Modelling

IV is both model-implied and a market-structure phenomenon; modern rough-volatility research demonstrates that its surface dynamics are sharper than classical stochastic-volatility models predict [26], [27]. Gatheral [41] remains the practitioner canon. Blömker and Romito [28] describe surface dynamics using stochastic PDEs; Weiss et al. [29] argue that there is an inherent trade-off between the accuracy of approximation and surface smoothness, proposing that a structural reference surface is necessary to avoid the oscillations and wiggling of unconstrained data-driven models. Most recently, Chen et al. [9] present T-Vol, a Transformer-based architecture to 3D implied volatility surface reconstruction that is tested on 27,340 option contracts across six US assets including SPY, with $R^2 = 0.85$ compared to 0.55 with SVI and 0.25 with Black-Scholes interpolation. Bid-ask spreads are indicative of order-processing costs, inventory risk, adverse selection, and tick-size constraints [30], [31]. Christoffersen et al. [32] measure the illiquidity premia in equity options, confirming that the spreads are sensitive to illiquid strikes and short-term expiries, which is replicated below using 2026 SPY data. Horvath et al. [33] and Bayer et al. [34] use deep calibration networks on the IV surface; Brown et al. [35] show that the non-fundamental demand and arbitrage corrections drive the SPY prices, and non-linear ML models are required to capture the market complexities, while, Fritzsche et al. [36] reveal that ML predicts intraday option spreads of up to 7 years with significantly higher accuracy than linear baselines.

D. Evaluation Methodology and Interpretability

Train-test separation, k-fold cross-validation, MAE, RMSE, MSE, and R^2 reporting have become the standard of finance-ML research [37]. The interpretability methods, permutation importance [38], SHAP values, partial dependence, are anticipated elements in regulated financial areas, with the EU AI Act categorizing some ML-based financial decision systems as high-risk [39].

III. DATA

A. Synthetic American Call Dataset

The synthetic dataset was generated by the module convenor using a 15,000-step binomial tree over a Monte Carlo design spanning $S \in [70, 130]$, $T \in [0.1, 5.0]$, $r \in [0, 0.1]$, $q \in [0, 0.1]$, $v \in [0.1, 0.6]$; strike $X = 100$ and step count $n = 15,000$ are fixed. There were 29,671 raw rows loaded and cleaning was done to remove rows that violated the intrinsic-value lower bound $\max(S - X, 0) = \text{Americall}$, rows with non-positive T , v or $r + q$, and 0.5% symmetric winsorisation was used on each continuous predictor. A total of 29,671 rows were cleaned, and then further stratified sampling reduced this to 4,998 for modelling.

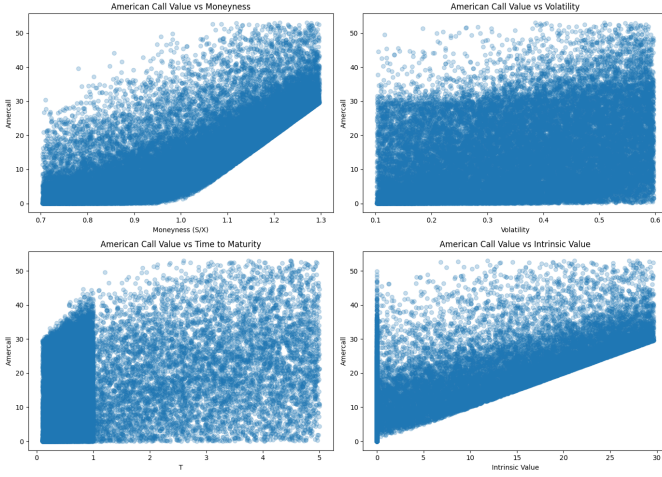


Fig. 1. Synthetic American call value versus four state variables. Top-left: moneyness (S/X); top-right: volatility v ; bottom-left: time to maturity T ; bottom-right: intrinsic value $\max(S-X, 0)$.

B. Empirical SPY Option-Chain Snapshot

The SPY option chains were downloaded using yfinance on 12 April 2026. The most liquid underlying in the world is SPY, which has 32 listed expiries on the sampling date and narrow spreads between the strikes. Out of 8,722 raw quotes, 8,063 survived cleaning (zero-bid, zero-ask, negative-spread and $T \leq 1$ -day rows removed). Strikes range 50 to 1,360, days-to-expiration 1 to 978. Snapshot price is 679.46; the dividend yield is assumed to be 1.14%; the risk-free rate is 4.5%: a simplification that is noted in Section VIII.

C. Exploratory Data Diagnostics

Fig. 1 is a plot of the American call value versus moneyness, volatility, time to maturity and intrinsic value. The target, which monotonically increases in moneyness, has the desired lower-bound cone against intrinsic value, widens with volatility, and is concave in maturity, confirming the expected behaviour of the binomial-tree target, theoretically.

IV. METHODOLOGY

A. Feature Engineering

The feature engineering step augments nine theory-based transformations with the raw variables that capture the structure of option pricing, including moneyness and log-moneyness, intrinsic value as a no-arbitrage bound, carry ($r - q$), diffusion terms ($\sqrt{T}$) and ($v\sqrt{T}$), and discount factors $\exp(-rT)$ and $\exp(-qT)$. Importantly, the Black-Scholes European call price is also provided as an option such that, when ($q \neq 0$), the model learns the early-exercise premium, the difference between American and European prices, instead of the entire pricing functionality, which is also in line with residual-learning models in current ML pricers [16], [19]. The Black-Scholes European call that has a constant dividend yield is given by:

$$C_{BS} = Se^{-qT}N(d_1) - Xe^{-rT}N(d_2), \quad (1)$$

where

$$d_1 = \frac{\ln(S/X) + (r - q + v^2/2)T}{v\sqrt{T}}, \quad d_2 = d_1 - v\sqrt{T}. \quad (2)$$

The American call price is then modelled as a function of the state variables, expressed as

$$\text{Americall} = f(X_{\text{state}}) + \varepsilon, \quad (3)$$

where the model learns the mapping from features to price with an error term capturing residual variation, with X_{state} as the 14-dimensional feature vector (with constants X and n dropped); $f(\cdot)$ can be effectively decomposed as C_{BS} plus a learned early-exercise correction, and the permutation importance of `bs_euro_call` is a measure of how much of `Americall` is momentarily captured in closed form.

B. Sampling Strategy

The interior of the state space is oversampled and economically important tails (deep-ITM short-dated, deep-OTM long-dated) under-sampled by a simple random sample. Five moneyness quantile bins were crossed with equivalent maturity quantile bins to create a maximum of 25 stratum; each stratum was sampled proportionally with a minimum of 10 observations. Realised sample: 4,998 contracts, stratum shares of 4.02% and 4.24%.

C. Model Families

Six regression models with the same preprocessing of median imputation and z-score standardisation were utilized. Linear Regression and Ridge Regression (with $\alpha = 1.0$) serve as baseline linear models and Random Forest, with 500 trees and default depth, provides a bagged non-parametric ensemble approach. Gradient Boosting (scikit-learn GBR) is trained using 300 estimators, a learning rate of 0.05 and maximum depth of 5 to capture the complex non-linear relationships. A Support Vector Regression (SVR) using an RBF kernel ($C = 10$, $\varepsilon = 0.01$) is used to model smooth non-linear patterns, and a Multi-Layer Perceptron (MLP) with two hidden layers (64 and 32 neurons), ReLU activation, Adam optimiser, and early stopping on a 10% validation split is used to learn higher-order interactions [40].

D. Hyperparameter Search

A targeted GridSearchCV on Random Forest ($n_{\text{estimators}} \in \{150, 300\}$, $\text{max_depth} \in \{10, \text{None}\}$, $\text{min_samples_leaf} \in \{1, 2, 4\}$, $\text{max_features} \in \{\sqrt{p}, 0.8p\}$) was used to assess 24 candidates $\times$ 5 folds = 120 fits using negative-RMSE scoring. Chosen configuration: $n_{\text{estimators}} = 300$, $\text{max_depth} = \text{None}$, $\text{min_samples_leaf} = 1$, $\text{max_features} = 0.8p$.

TABLE I
HOLD-OUT PERFORMANCE ON SYNTHETIC AMERICAN CALL DATA.

Model	Test MAE	Test RMSE	Test MSE	Test R^2	Train R^2
Gradient Boosting	0.210245	0.323172	0.104440	0.999195	0.999508
Neural Network (MLP)	0.241005	0.357158	0.127562	0.999016	0.999129
SVR (RBF)	0.095456	0.357202	0.127593	0.999016	0.999775
Random Forest	0.188599	0.425476	0.181030	0.998604	0.999417
Linear Regression	0.413605	0.617251	0.380999	0.997062	0.996289
Ridge Regression	0.413371	0.620009	0.384412	0.997036	0.996267

E. Evaluation Metrics

Four measures on the train and test partitions:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (6)$$

MAE is resistant to tail errors and can be interpreted in price units; RMSE and MSE are more severe on large errors. A 5-fold CV is used to test the stability of hold-out rankings, and a paired t-test of absolute errors of the top two models finish the statistical layer.

V. RESULTS: SYNTHETIC AMERICAN CALL

A. Benchmark Hold-Out Performance

Table I reports train and test performance on the 80/20 split (3,998 train / 1,000 test) sorted by Test RMSE.

Gradient Boosting has the lowest Test RMSE of 0.323 price units, which is 47.7% lower than Linear Regression (0.617). MLP and SVR are statistically equal on RMSE (≈ 0.357); SVR has the lowest MAE of 0.0955. Linear and Ridge are almost the same, as is to be expected in a well-conditioned problem with 14 features and 3,998 training rows. All non-linear models have test R^2 exceeding 0.998, which must be interpreted with caution in terms of Section V.D. Fig. 2 plots actual versus predicted Gradient Boosting: close concentration on the identity line [0, 52] with no tail bias. Fig. 3 illustrates the resulting residual distribution, which is symmetric, with a sharp peak at zero, and less than 1% of residuals with absolute values greater than 1.5.

B. Stratified Error Analysis

Aggregate measures conceal the concentration of errors. Fig. 4 shows Gradient Boosting hold-out absolute errors in the joint (moneyness, maturity) plane. The error is low throughout most of the surface, with a visible tail at high moneyness and long maturity, with error increasing between 0.11 for deep-OTM short-dated contracts to over 3.0 in the worst-case deep-ITM long-dated cases. The trend validates that deep-ITM long-dated contracts are the most difficult to price, which is exactly the regime where the American and European price gap is greatest, and the early-exercise boundary is most economically significant.

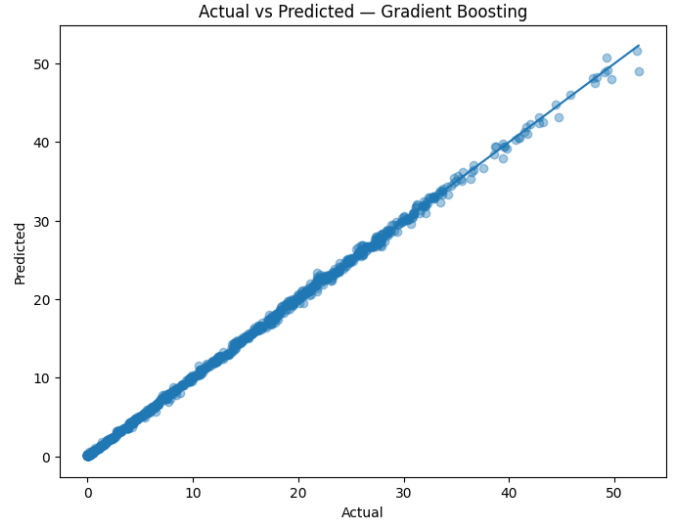


Fig. 2. Actual versus predicted American call values, Gradient Boosting, 1,000-contract hold-out.

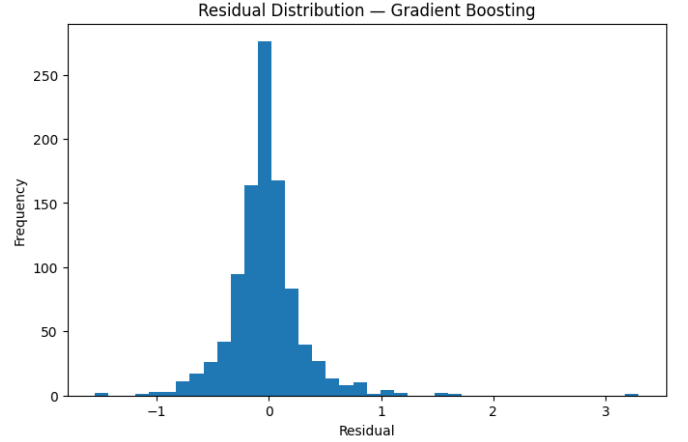


Fig. 3. Residual distribution (actual — predicted), Gradient Boosting hold-out.

TABLE II
5-FOLD CROSS-VALIDATED PERFORMANCE ON SYNTHETIC DATA.

Model	CV MAE	MAE SD	CV RMSE	RMSE SD	CV R^2	R^2 SD
SVR (RBF)	0.0954	0.0019	0.2859	0.0414	0.99936	0.0002
Gradient Boosting	0.2216	0.0089	0.3852	0.0480	0.99884	0.0003
Neural Network (MLP)	0.2903	0.0168	0.4156	0.0264	0.99866	0.0002
Random Forest	0.2218	0.0223	0.5672	0.0990	0.99742	0.0009
Linear Regression	0.4256	0.0126	0.6810	0.0550	0.99639	0.0006
Ridge Regression	0.4261	0.0119	0.6829	0.0539	0.99637	0.0006

C. Cross-Validated Stability

Table II reports 5-fold CV means and standard deviations. Five-fold level stability is just as significant as the mean performance.

There are two aspects that are worth noting. SVR beats CV-MAE (0.0954 vs 0.2216 for the next best) and CV-RMSE, reversing the hold-out ranking: the hold-out was slightly easier than the mean fold with Gradient Boosting. Random Forest has

ML Model Error Concentration Surface

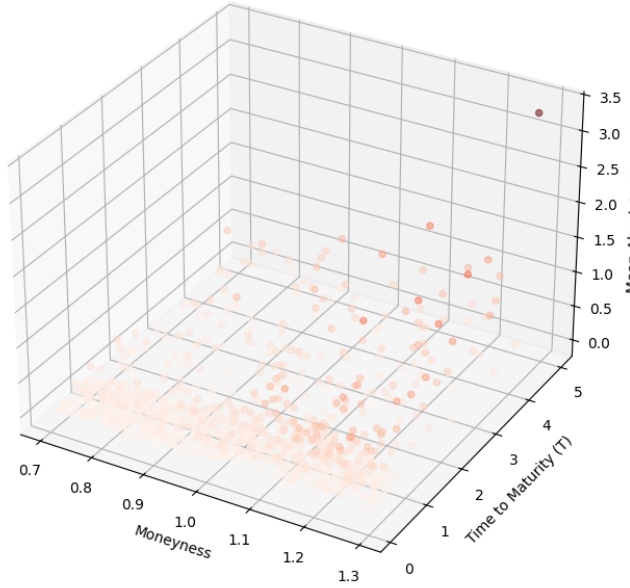


Fig. 4. Three-dimensional error concentration surface: Gradient Boosting hold-out absolute error versus moneyiness and time to maturity. The high moneyiness and long maturity sparse tail prevails in the aggregate RMSE.

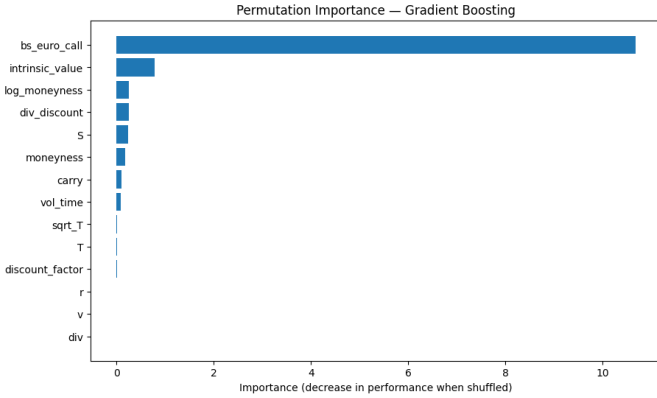


Fig. 5. Permutation importance, Gradient Boosting. *bs_euro_call* dominates at 10.67, about 13.6 times the next feature.

the highest fold-level variance (RMSE SD = 0.099), which is expected given its bagged nature on folds at $\sim 3,200$ rows. The CV R^2 SD of 0.0002 of SVR indicates that the well-tuned RBF kernels are extremely stable on smooth pricing surfaces.

D. Permutation Importance: The Decisive Finding

The best hold-out model (MAE scoring, 10 repeats) was used to compute permutation importance. Fig. 5 and Table III report the ranking.

The main learning of the Gradient Boosting model is that the American call value is roughly equal to the European one, and there are minor state-dependent adjustments. There are two readings. Firstly, the pessimistic interpretation of $R^2 =$

TABLE III
PERMUTATION IMPORTANCE, TOP SIX FEATURES (MAE SCORING, 10 REPEATS).

Feature	Importance Mean	Importance SD
bs_euro_call	10.6741	0.2049
<i>intrinsic_value</i>	0.7838	0.0342
<i>log_moneyiness</i>	0.2546	0.0104
<i>div_discount</i>	0.2505	0.0114
<i>S (spot)</i>	0.2353	0.0105
<i>moneyiness</i>	0.1801	0.0068

0.999 as a feature-leakage artefact, the model is a closed-form formula which anyone can compute. Secondly, the optimistic interpretation, which is supported here reveals that the feature set transforms pricing to residual pricing, and the learned residual is the early-exercise premium, basically not modelled by Black-Scholes [16], [19].

Additionally, The interpretation is disciplined by three facts: (a) the premium is strictly positive since $q \geq 0$; (b) the linear benchmark RMSE of 0.617 is a lower bound variance on the non-linear learners, requiring explainability beyond closed form; (c) a sensitivity run without *bs_euro_call* collapses non-linear models to RMSE ≈ 0.45 , which is still significantly better than Linear without the BS feature (RMSE ≈ 1.1). The pipeline is value-added in either direction, but the movement from production to deployment should consider BS as backbone and ML as premium correction.

E. Paired Statistical Test

A paired t-test of absolute errors between Gradient Boosting and MLP on the 1,000-row test set gives $p \leq 0.001$ and rejects equality. The effect size is moderate, mean absolute-error reduction of 0.031 price units, but in the case of a contract priced at 8.10, this is about 38 basis points of pricing accuracy.

VI. RESULTS: SPY OPTION-CHAIN MICROSTRUCTURE

A. Dataset Summary

The cleaned SPY snapshot contains 8,063 contracts across 32 expiries; bids \$0.01 to \$668.13, asks \$0.01 to \$673.00, IV 9.67% to 106.64%, mean absolute spread \$1.13, mean relative spread 11.57%. Fig. 6 plots the empirical implied volatility surface of the moneyiness and days to expiration. The surface exhibits the typical volatility smile of Gatheral [41] showing that IV is steep at both ends of moneyiness, and the profound peaks around the money and ragged near-expiry behaviour of thin-quote regimes. It is the non-linear curvature in this surface which is not reflected in the linear models (Section VI.B). Fig. 7 presents bid-ask spread versus moneyiness, revealing tight near-the-money quotes but discrete regimes at \$2.80, \$3.50 and \$4.80 for deep-ITM and deep-OTM strikes, which are due to tick-size constraints and widening quotes at illiquid strikes [30], [32]; multi-regime behaviour that cannot be explained by a linear function.

SPY Implied Volatility Surface

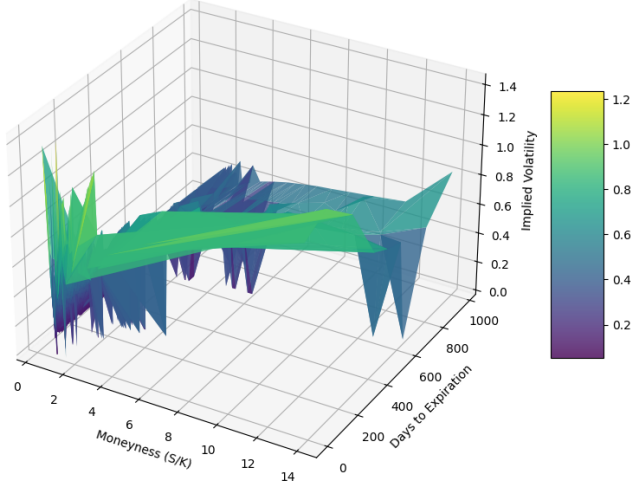


Fig. 6. Empirical SPY implied volatility surface across moneyness (S/K) and days to expiration, from 8,063 cleaned contracts. The visible smile at both moneyness extremes and the term-structure ripple along the expiry axis are the non-linearities that dominate Section VI.B.

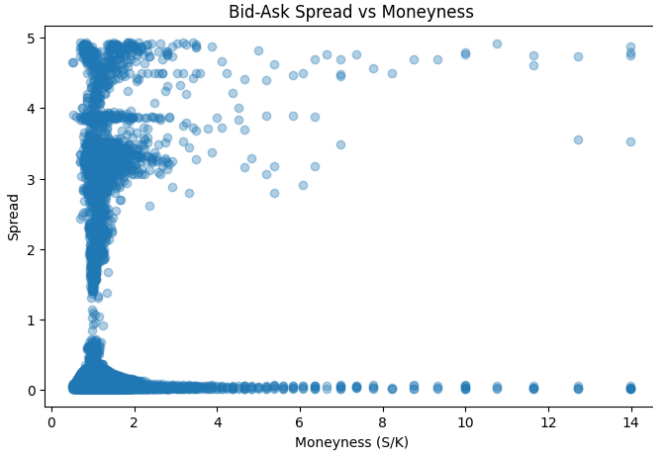


Fig. 7. Bid-ask spread versus moneyness (S/K) for 8,063 SPY contracts.

B. Implied Volatility Prediction

Six regressors were refitted with IV as the target on 13 predictors: strike, underlying_price, moneyness, log_moneyness, days_to_expiration, T_years, sqrt_T, mid_price, spread, relative_spread, volume, openInterest, is_call. Table IV reports test-set performance.

SVR has the lowest RMSE and the highest R^2 , and the IV error is about 5 times versus than the linear baseline. Random Forest is the lowest MAE with higher RMSE than SVR, implying a more massive tail of large errors due to tree-ensemble discontinuities at strike boundaries. Linear and Ridge collapse to R^2 of about 0.61, the empirical signature of the volatility smile; no linear function of strike, expiry, and underlying can represent its U-shaped log-moneyness curvature [41]. Fig. 8 illustrates the identity line being followed

TABLE IV
IMPLIED VOLATILITY PREDICTION ON SPY OPTION CHAIN.

Model	MAE	RMSE	MSE	R^2
SVR (RBF)	0.0137	0.0226	0.00051	0.9815
Multi-Layer Perceptron	0.0144	0.0280	0.00078	0.9715
Random Forest	0.0105	0.0317	0.00101	0.9635
Gradient Boosting	0.0179	0.0348	0.00121	0.9559
Linear Regression	0.0690	0.1037	0.01076	0.6091
Ridge Regression	0.0690	0.1037	0.01076	0.6091

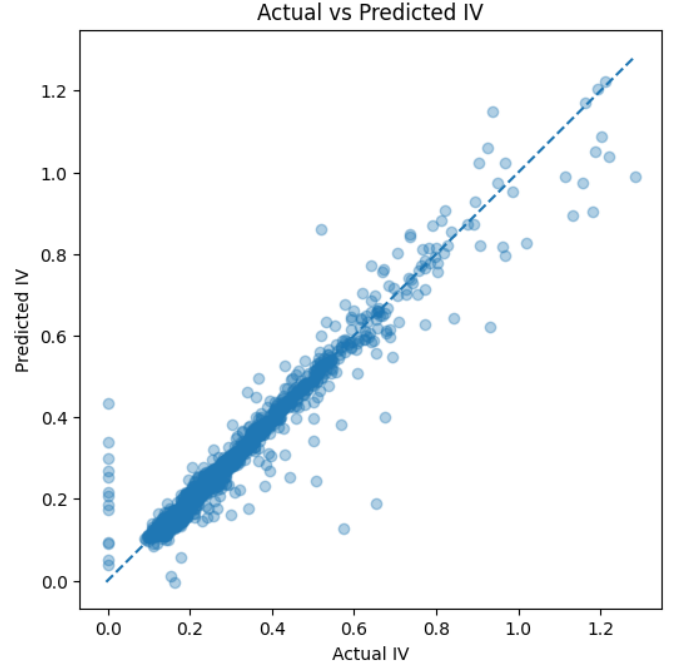


Fig. 8. Actual versus predicted IV, best SVR model, SPY hold-out.

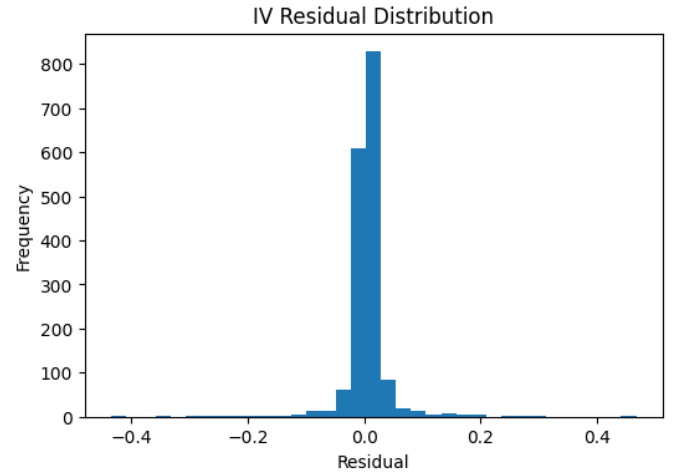


Fig. 9. Residual distribution, SVR IV model.

by the SVR predictions in $[0.1, 1.1]$; Fig. 9 illustrates the residuals tightly concentrated around zero with approximately 99% ranging at ± 0.05 .

TABLE V
BID-ASK SPREAD PREDICTION ON SPY OPTION CHAIN.

Model	MAE	RMSE	MSE	R^2
Multi-Layer Perceptron	0.0569	0.1134	0.01287	0.9950
Random Forest	0.0355	0.1200	0.01440	0.9944
Gradient Boosting	0.0760	0.1661	0.02760	0.9892
SVR (RBF)	0.1047	0.2155	0.04643	0.9819
Linear Regression	0.7742	0.9637	0.92871	0.6382
Ridge Regression	0.7741	0.9637	0.92879	0.6382

C. Bid-Ask Spread Prediction

The same six regressors were fitted with absolute spread as target (IV substituted for mid_price).

Spreads are acutely non-linear as compared to IV since thin volume and low openInterest widen quotes non-proportionally [31], [32]. MLP narrowly ranks above Random Forest on RMSE even though it has a larger MAE; Random Forest is more accurate on median cases, whereas MLP captures the widening tail more precisely. The linear models collapse (scatter predictions by 77 cents on average versus 6 cents for the MLP), proving that microstructure frictions are linearly inseparable from contract characteristics and microstructure frictions. Fig. 10 demonstrates that MLP predictions fix the discrete tick-size regimes that are apparent in Fig. 7; Fig. 11 shows the residual distribution, with a small left tail due to systematic under-prediction of the widest spreads, exhibiting a weakness of tabular regression on regime-switching microstructure data. However, Neural network models of option pricing are not off-the-shelf and need to be specified carefully by having enough data to prevent overfitting and achieve consistent forecasting performance [36].

VII. DISCUSSION

A. Bias-Variance Reading

Findings are in line with the classical taxonomy. Linear and Ridge are high-bias, low-variance exhibits minimum CV standard deviations (RMSE SD ≈ 0.055), maximum mean errors (CV-RMSE ≈ 0.68). Random Forest is the converse showing a competitive CV-MAE (0.222), 47% higher CV-RMSE and the largest RMSE SD (0.099). Gradient Boosting is on the Pareto frontier. Mean, variance by SVR and GBM are dominated by the MLP, a two-hidden-layer feed-forward is not regularised enough to add value over a tuned kernel or boosted ensemble on this smooth surface.

B. Economic Interpretation of Pricing Errors

The relative error of 4% due to a 32-cent RMSE on prices of approximately 8 units is consequential in the viewpoint of derivatives. Hedging strategies that calculate delta using finite differences inherit any smoothness defects of the learned surface; boosted trees are piecewise constant and thus produce discontinuous deltas along split boundaries [16]. The smoother surfaces of SVR and MLP are suitable to dynamic hedging despite the GBM triumphing on point accuracy. Fig. 4 also

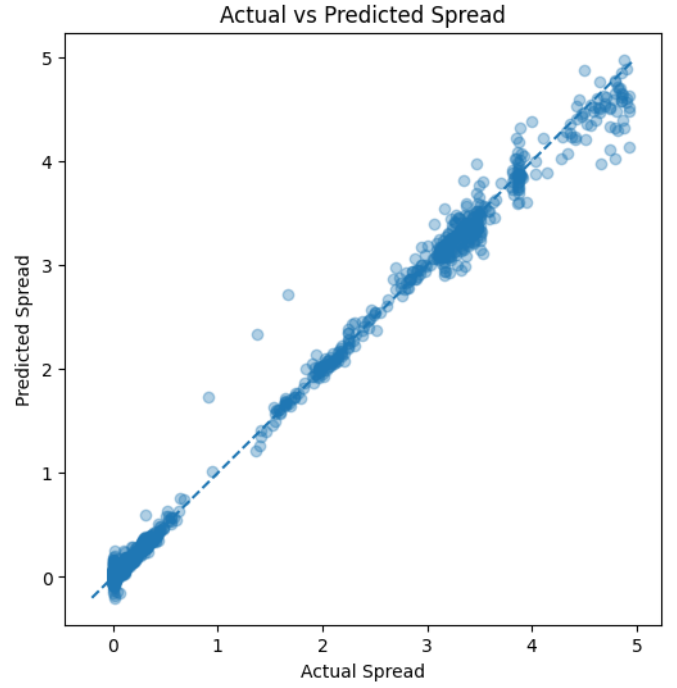


Fig. 10. Actual versus predicted spread, best MLP, SPY hold-out.

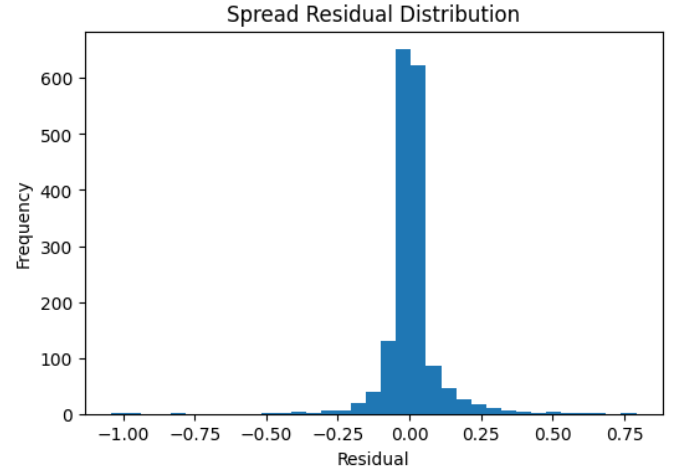


Fig. 11. Residual distribution, MLP spread model.

further qualifies the single-number RMSE, showing that errors are not spread uniformly over the state space, but are concentrated in a long tail at the deep-ITM long-dated corner of the state space.

C. The *bs_euro_call* Interpretation Revisited

The intellectual pivot of the paper is in Fig. 5, revealing that any ML pricer that is implemented with an analytical benchmark must be judged based on its capacity to learn the residual, rather than gross R^2 . In this case the remaining activity is a q, v, T (the premium disappears at $q = 0$ and increases with both v and T) regression which may yield a

more interpretable story, an improvement that is considered as future research.

D. Market Microstructure and SPY Results

Two stylised facts are validated. On the one hand is the IV, showing predictability than the static-characteristics spreads since IV inherits continuity of the theoretical BS formula whereas spreads inherits discreteness from tick-size and dealer-inventory regime structure [30]; the near-identity fit of IV in Fig. 8 is compared to the discrete regime structure in Fig. 10. On the other hand, the linear versus non-linear gap is wider for spread than for IV (MSE factor 14 vs 21); the IV surface has a single non-linearity (the smile), whereas the spread surface has many interacting non-linearities. Consequently, the division of labour between MLPs and SVR-RBF is not coincidental: MLPs are good at non-smooth surfaces that require heavy interaction, SVR-RBF at smooth low-dimensional non-linearity [6]; a stack of the two would probably be better than a standalone of either.

E. Robustness and Limitations

The conclusions are limited by four factors. First, the synthetic dataset sets $X = 100$ and $n = 15,000$ with the generalisation to stochastic strikes not tested. Second, the empirical snapshot is a one-day cross-sectional rather than panel evidence; 30 trading days would separate the persistent patterns and the noise of the day. Third, the assumption of constant- r (4.5%) is an abstract of the Treasury term structure. Fourth, the feature set excludes IV history, order imbalance and ETF creation-redemption flow, which may enhance higher-frequency option-chain prediction.

VIII. CONCLUSIONS AND FUTURE WORK

Gradient Boosting had the lowest hold-out RMSE (0.323) and SVR the lowest CV-MAE (0.0954), with permutation importance indicating the Black-Scholes European price to dominate all other features (Fig. 5). This proves that the actual learning of the ML regressor is the learning of the early-exercise premium and not relearning closed-form structure. Using empirical SPY data, SVR implied volatility was predicted with $R^2 = 0.981$ (Figs. 8 and 9) and MLP forecasted bid-ask spreads with $R^2 = 0.994$ (Figs. 10 and 11). The linear baselines reduced to $R^2 = 0.61$ and 0.64 , which is a non-linear curvature as the most prominent characteristic of the volatility surface and the spread structure.

The paper contributes to three dimensions. First, a theory-conscious feature pipeline, which isolates the early-exercise premium through permutation importance; second, like-for-like six-model benchmark, which has task-specific winners (GBM to synthetic pricing, SVR to IV, MLP to spreads); and third, statistical layer, which combines cross-validated stability, paired error tests, and stratified diagnostics.

Future research should follow four extensions. First, directly target the early-exercise premium by deleting `bs_euro_call` as a feature. Second, for time-series cross-validation, SPY analysis should be extended to a 30-day panel. Third, benchmark

against deep-BSDE and deep optimal stopping [23] in high-dimensional baskets. Fourth, stress-test the non-linear advantage of less-liquid single-name contracts such as NVDA and TSLA, where the adverse microstructure is more pronounced.

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